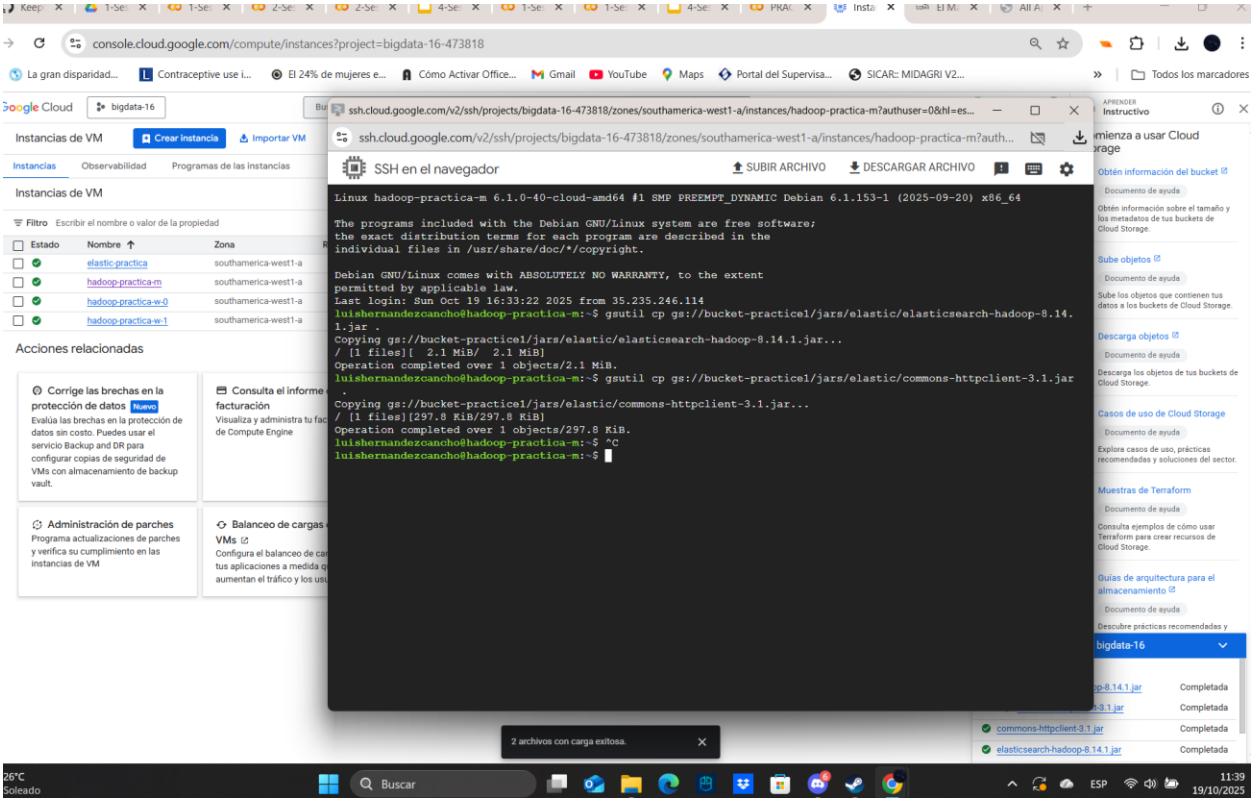
PRACTICA BIG DATA ARCHITECTURE

1. Tarea 1:



2. Tarea 2:

ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/elastic-practica?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true

SSH en el navegador

SUBIR ARCHIVO DESCARGAR ARCHIVO

```
Linux elastic-practica 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Oct 19 17:22:53 2025 from 35.235.246.113
luishernandezcancho@elastic-practica:~$ sudo cat /etc/elasticsearch/elasticsearch.yml
# ======================== Elasticsearch Configuration =========================
#
# NOTE: Elasticsearch comes with reasonable defaults for most settings.
#       Before you set out to tweak and tune the configuration, make sure you
#       understand what are you trying to accomplish and the consequences.
#
# The primary way of configuring a node is via this file. This template lists
# the most important settings you may want to configure for a production cluster.
#
# Please consult the documentation for further information on configuration options:
# https://www.elastic.co/guide/en/elasticsearch/reference/index.html
#
# ---------------------------------- Cluster -----------------------------------
#
# Use a descriptive name for your cluster:
#
#cluster.name: my-application
#
# ------------------------------------ Node ------------------------------------
#
# Use a descriptive name for the node:
#
#node.name: node-1
#
# Add custom attributes to the node:
#
#node.attr.rack: r1
#
# ---------------------------------- Paths -------------------------------------
#
# Path to directory where to store the data (separate multiple locations by comma):
#
path.data: /var/lib/elasticsearch
#
# Path to log files:
#
path.logs: /var/log/elasticsearch
#
# ---------------------------------- Memory ------------------------------------
#
# Lock the memory on startup:
#
```

USD/PEN 12:28

ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/elastic-practica?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true - Google Chrome

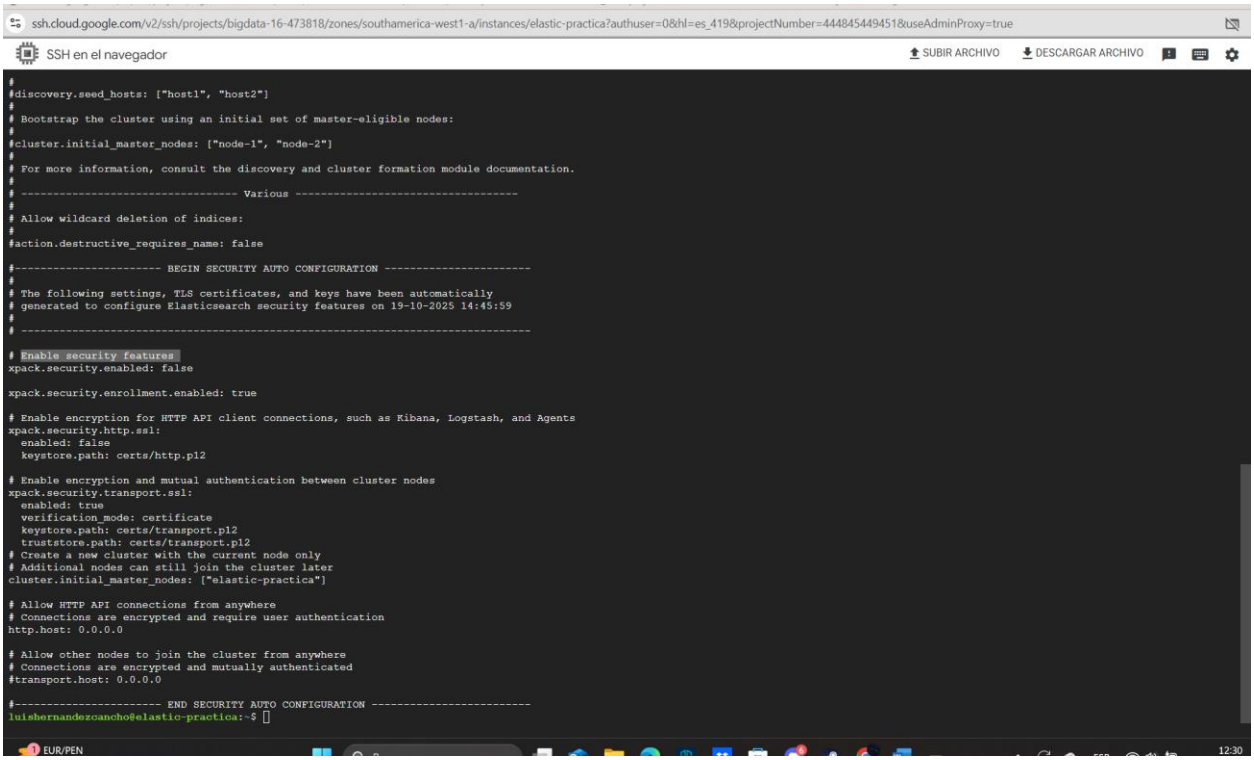
ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/elastic-practica?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true

SSH en el navegador

SUBIR ARCHIVO DESCARGAR ARCHIVO

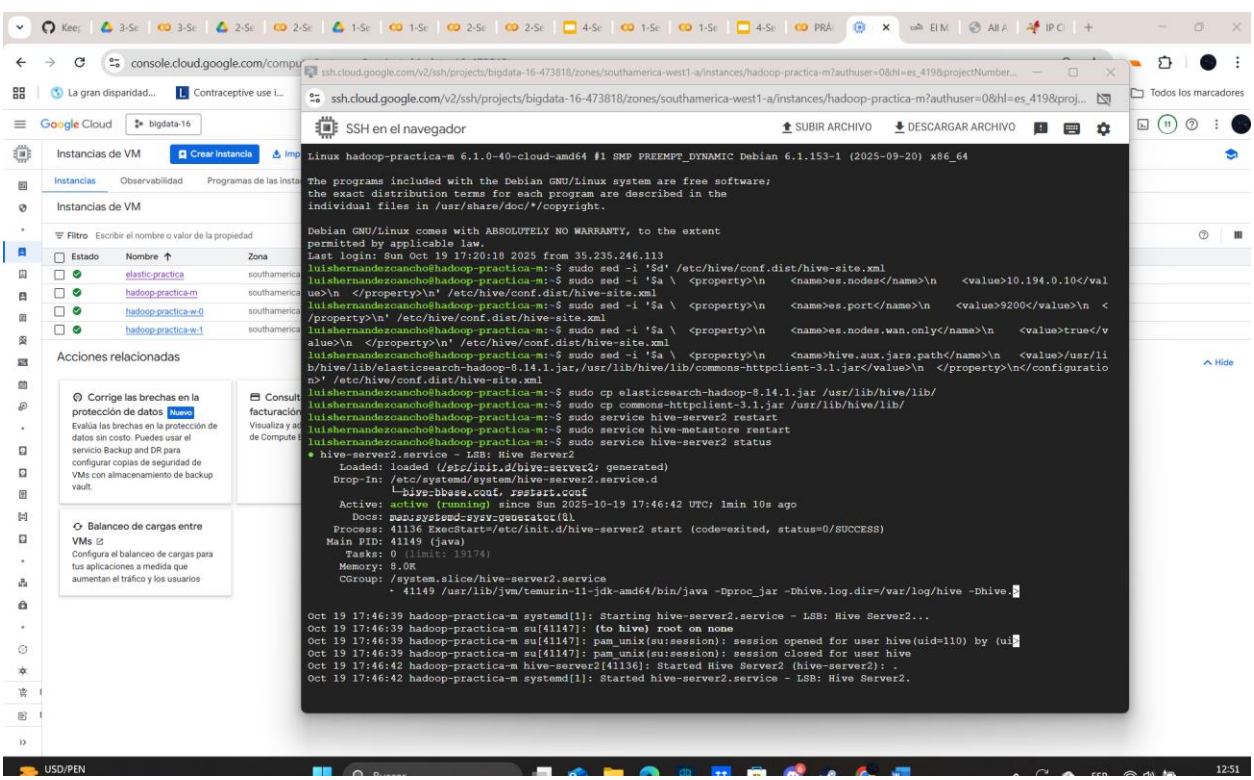
```
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ---------------------------------- Network -----------------------------------
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
#network.host: 192.168.0.1
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
#http.port: 9200
#
# For more information, consult the network module documentation.
#
# --------------------------------- Discovery ----------------------------------
#
# Pass an initial list of hosts to perform discovery when this node is started:
# The default list of hosts is ["127.0.0.1", "[::1]"]
#
#discovery.seed_hosts: ["host1", "host2"]
#
# Bootstrap the cluster using an initial set of master-eligible nodes:
#
#cluster.initial_master_nodes: ["node-1", "node-2"]
#
# For more information, consult the discovery and cluster formation module documentation.
#
# ---------------------------------- Various -----------------------------------
#
# Allow wildcard deletion of indices:
#
#action.destructive_requires_name: false

#----------------------- BEGIN SECURITY AUTO CONFIGURATION -----------------------
#
# The following settings, TLS certificates, and keys have been automatically
# generated to configure Elasticsearch security features on 19-10-2025 14:45:59
#
# -------------------------------------------------------------------------------

# Enable security features
xpack.security.enabled: false
```

EUR/PEN 12:30


ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/elastic-practica?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true
SSH en el navegador
SUBIR ARCHIVO
DESCARGAR ARCHIVO
#
#discovery.seed_hosts: ["host1", "host2"]
#
Bootstrap the cluster using an initial set of master-eligible nodes:
#
#cluster.initial_master_nodes: ["node-1", "node-2"]
#
For more information, consult the discovery and cluster formation module documentation.
#
---------------------------------- Various -----------------------------------
#
Allow wildcard deletion of indices:
#
#action.destructive_requires_name: false
#---------------------- BEGIN SECURITY AUTO CONFIGURATION ----------------------
#
The following settings, TLS certificates, and keys have been automatically
generated to configure Elasticsearch security features on 19-10-2025 14:45:59
#

Enable security features
xpack.security.enabled: false
xpack.security.enrollment.enabled: true
Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
enabled: false
keystore.path: certs/http.p12
Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
enabled: true
verification_mode: certificate
keystore.path: certs/transport.p12
truststore.path: certs/transport.p12
Create a new cluster with the current node only
Additional nodes can still join the cluster later
cluster.initial_master_nodes: ["elastic-practica"]
Allow HTTP API connections from anywhere
Connections are encrypted and require user authentication
http.host: 0.0.0.0
Allow other nodes to join the cluster from anywhere
Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0
#---------------------- END SECURITY AUTO CONFIGURATION ------------------------
luishernandezcancho@elastic-practica:~$
EUR/PEN
12:30

3. Tarea 3:


console.cloud.google.com/compu
Google Cloud
bigdata-16
Instancias de VM
Crear instancia
Instancias
Observabilidad
Filtro
Escribir el nombre o valor de la propiedad
Estado
Nombre
Zona
elastic-practica
hadoop-practica-m
hadoop-practica-w-0
hadoop-practica-w-1
Acciones relacionadas
Corrige las brechas en la protección de datos
Nuevo
Evalúa las brechas en la protección de datos sin costo. Puedes usar el servicio Backup and DR para configurar copias de seguridad de VMs con almacenamiento de backup vault.
Balanceo de cargas entre VMs
Configura el balanceo de cargas para tus aplicaciones a medida que aumentan el tráfico y los usuarios
ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/hadoop-practica-m?authuser=0&hl=es_419&projectNumber...
SSH en el navegador
SUBIR ARCHIVO
DESCARGAR ARCHIVO
Linux hadoop-practica-m 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Oct 19 17:20:18 2025 from 35.235.246.113
luishernandezcancho@hadoop-practica-m:~$ sudo sed -i 'Sd' /etc/hive/conf.dist/hive-site.xml
luishernandezcancho@hadoop-practica-m:~$ sudo sed -i '$a \ <property>\n <name>es.nodes</name>\n <value>10.194.0.10</value>\n </property>\n' /etc/hive/conf.dist/hive-site.xml
luishernandezcancho@hadoop-practica-m:~$ sudo sed -i '$a \ <property>\n <name>es.port</name>\n <value>9200</value>\n </property>\n' /etc/hive/conf.dist/hive-site.xml
luishernandezcancho@hadoop-practica-m:~$ sudo sed -i '$a \ <property>\n <name>es.nodes.wan.only</name>\n <value>true</value>\n </property>\n' /etc/hive/conf.dist/hive-site.xml
luishernandezcancho@hadoop-practica-m:~$ sudo sed -i '$a \ <property>\n <name>hive.aux.jars.path</name>\n <value>/usr/lib/hive/lib/elasticsearch-hadoop-8.14.1.jar,/usr/lib/hive/lib/commons-httpclient-3.1.jar</value>\n </property>\n</configuration>' /etc/hive/conf.dist/hive-site.xml
luishernandezcancho@hadoop-practica-m:~$ sudo cp elasticsearch-hadoop-8.14.1.jar /usr/lib/hive/lib/
luishernandezcancho@hadoop-practica-m:~$ sudo cp commons-httpclient-3.1.jar /usr/lib/hive/lib/
luishernandezcancho@hadoop-practica-m:~$ sudo service hive-server2 restart
luishernandezcancho@hadoop-practica-m:~$ sudo service hive-metastore restart
luishernandezcancho@hadoop-practica-m:~$ sudo service hive-server2 status
● hive-server2.service - LSB: Hive Server2
Loaded: loaded (/etc/init.d/hive-server2; generated)
Drop-In: /etc/systemd/system/hive-server2.service.d
└─hive-hbase.conf, restart.conf
Active: active (running) since Sun 2025-10-19 17:46:42 UTC; 1min 10s ago
Docs: man:systemd-sysv-generator(8)
Process: 41136 ExecStart=/etc/init.d/hive-server2 start (code=exited, status=0/SUCCESS)
Main PID: 41149 (java)
Tasks: 0 (limit: 19174)
Memory: 8.0K
CGroup: /system.slice/hive-server2.service
└─ 41149 /usr/lib/jvm/temurin-11-jdk-amd64/bin/java -Dproc_jar -Dhive.log.dir=/var/log/hive -Dhive.
Oct 19 17:46:39 hadoop-practica-m systemd[1]: Starting hive-server2.service - LSB: Hive Server2...
Oct 19 17:46:39 hadoop-practica-m su[41147]: (to hive) root on none
Oct 19 17:46:39 hadoop-practica-m su[41147]: pam_unix(su:session): session opened for user hive(uid=110) by (uid
Oct 19 17:46:39 hadoop-practica-m su[41147]: pam_unix(su:session): session closed for user hive
Oct 19 17:46:42 hadoop-practica-m hive-server2[41136]: Started Hive Server2 (hive-server2): .
Oct 19 17:46:42 hadoop-practica-m systemd[1]: Started hive-server2.service - LSB: Hive Server2.
USD/PEN
12:51

4. Tarea 4:

ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/hadoop-practica-m?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
Linux hadoop-practica-m 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
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permitted by applicable law.
Last login: Sun Oct 19 17:33:03 2025 from 35.235.249.97
luishernandezcancho@hadoop-practica-m:~$ curl -X POST "10.194.0.10:9200/_bulk" -H 'Content-Type: application/json' -d' { "index": { "_index": "alumnos", "_id": "3" } } { "id": 3, "name": "Carl
os", "last_name": "González" } { "index": { "_index": "alumnos", "_id": "4" } } { "id": 4, "name": "María", "last_name": "López" } { "index": { "_index": "alumnos", "_id": "5" } } { "id": 5, "
name": "Luis", "last_name": "Martínez" } { "index": { "_index": "alumnos", "_id": "7" } } { "id": 7, "name": "Sofía", "last_name": "Ramírez" } { "index": { "_index": "alumnos", "_id": "8" } }
{ "id": 8, "name": "Pedro", "last_name": "Hernández" } '
{"error":{"root_cause":[{"type":"illegal_argument_exception","reason":"The bulk request must be terminated by a newline [\\n]"}],"type":"illegal_argument_exception","reason":"The bulk request
must be terminated by a newline luishernandezcancho@hadoop-practica-m:~$ curl -X POST "10.194.0.10:9200/_bulk" -H 'Content-Type: application/json' -d 'application/json' -d '
{ "index": { "_index": "alumnos", "_id": "3" } }
{ "id": 3, "name": "Carlos", "last_name": "González" }
{ "index": { "_index": "alumnos", "_id": "4" } }
{ "id": 4, "name": "María", "last_name": "López" }
{ "index": { "_index": "alumnos", "_id": "5" } }
{ "id": 5, "name": "Luis", "last_name": "Martínez" }
{ "index": { "_index": "alumnos", "_id": "7" } }
{ "id": 7, "name": "Sofía", "last_name": "Ramírez" }
{ "index": { "_index": "alumnos", "_id": "8" } }
{ "id": 8, "name": "Pedro", "last_name": "Hernández" }
'
{"errors":false,"took":84,"items":[{"index":{"_index":"alumnos","_id":"3","_version":1,"result":"created","_shar                                    ds":{"total":2,"successful":1,"failed":0},"_seq_n
o":1,"_primary_term":1,"status":201}},{"index":{"_index":"alumn                                os","_id":"4","_version":1,"result":"created","_shards":{"total":2,"successful":1,"failed":0},"_se
q_no":2,"_prim                                ary_term":1,"status":201}},{"index":{"_index":"alumnos","_id":"5","_version":1,"result":"created","_shards":{"to                                    tal"
:2,"successful":1,"failed":0},"_seq_no":3,"_primary_term":1,"status":201}},{"index":{"_index":"alumnos","_id                                    ":"7","_version":1,"result":"created","_shards":{"tot
al":2,"successful":1,"failed":0},"_seq_no":4,"_primary_term                                    ":1,"status":201}},{"index":{"_index":"alumnos","_id":"8","_version":1,"result":"created","_shards":{"
total":2,"                                successful":1,"failed":0},"_seq_no":5,"_primary_term":1,"status":201}}]}luishernandezcancho@hadoop-practica-m:~$
luishernandezcancho@hadoop-practica-m:~$ curl -X GET "http://10.194.0.10:9200/alumnos/_search?pretty"
{
  "took" : 9,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 6,
      "relation" : "eq"
    },
    "max_score" : 1.0,
    "hits" : [
      {
        "_index" : "alumnos",
```
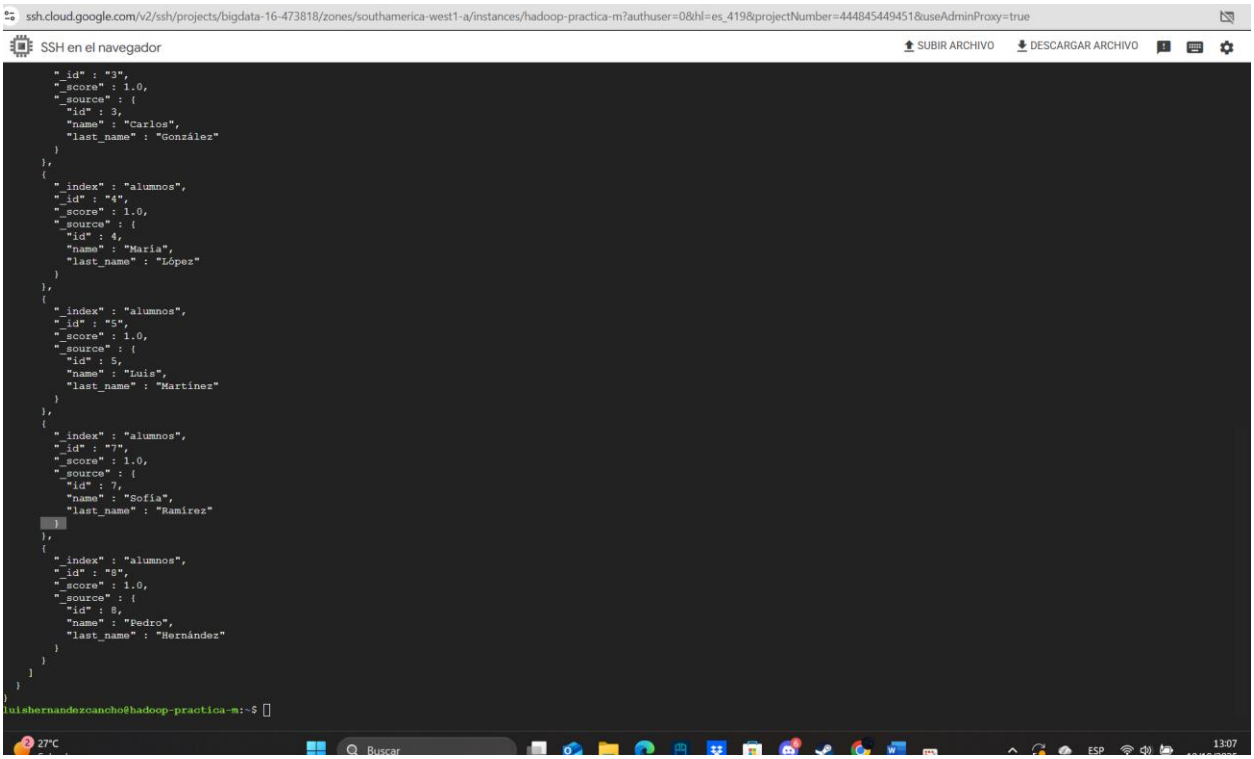
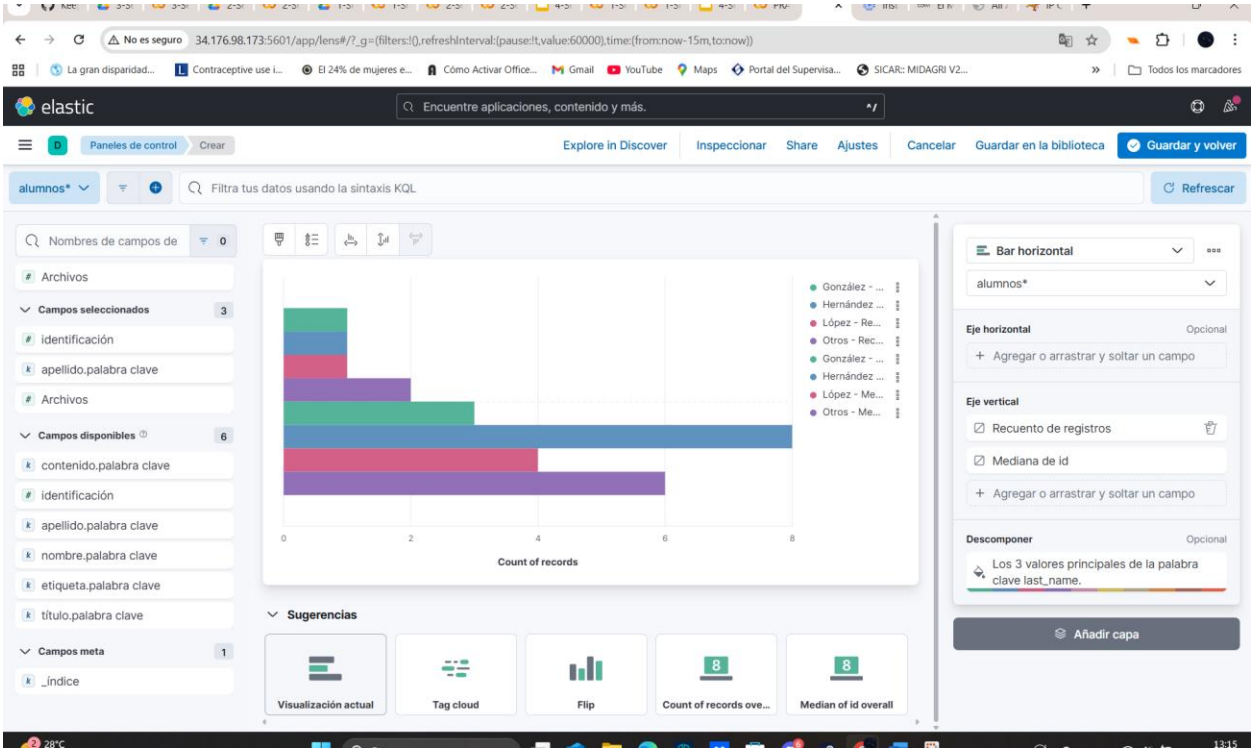

ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/hadoop-practica-m?authuser=0&hl=es_419&projectNumber=444845449451&useAdminProxy=true

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
      {
        "_index" : "alumnos",
        "_id" : "6",
        "_score" : 1.0,
        "_source" : {
          "title" : "New Document",
          "content" : "This is a new document for the master class",
          "tag" : [
            "general",
            "testing"
          ]
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "3",
        "_score" : 1.0,
        "_source" : {
          "id" : 3,
          "name" : "Carlos",
          "last_name" : "González"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "4",
        "_score" : 1.0,
        "_source" : {
          "id" : 4,
          "name" : "María",
          "last_name" : "López"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "5",
        "_score" : 1.0,
        "_source" : {
          "id" : 5,
          "name" : "Luis",
          "last_name" : "Martínez"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "7",
        "_score" : 1.0,
        "_source" : {
          "id" : 7,
          "name" : "Sofía",
          "last_name" : "Ramírez"
        }
```


ssh.cloud.google.com/v2/ssh/projects/bigdata-16-473818/zones/southamerica-west1-a/instances/hadoop-practica-m?authuser=0&hl=es_419&projectNumber=4448454494518useAdminProxy=true
SSH en el navegador
SUBIR ARCHIVO
DESCARGAR ARCHIVO
luishernandezcancho@hadoop-practica-m:~$

5. Tarea 5:


No es seguro
34.176.98.173:5601/app/lens#/?_g=(filters:!(),refreshInterval:(pause:!t,value:60000),time:(from:now-15m,to:now))
elastic
Encuentre aplicaciones, contenido y más.
Paneles de control
Crear
Explore in Discover
Inspeccionar
Share
Ajustes
Cancelar
Guardar en la biblioteca
Guardar y volver
alumnos*
Filtra tus datos usando la sintaxis KQL
Refrescar
Nombres de campos de
Archivos
Campos seleccionados
identificación
apellido.palabra clave
Archivos
Campos disponibles
contenido.palabra clave
identificación
apellido.palabra clave
nombre.palabra clave
etiqueta.palabra clave
título.palabra clave
Campos meta
_índice
Count of records
Sugerencias
Visualización actual
Tag cloud
Flip
Count of records ove...
Median of id overall
Bar horizontal
alumnos*
Eje horizontal
Agregar o arrastrar y soltar un campo
Eje vertical
Recuento de registros
Mediana de id
Agregar o arrastrar y soltar un campo
Descomponer
Los 3 valores principales de la palabra clave last_name.
Añadir capa